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REQUEST FOR DISBURSMENT UNDER THE RAPID CREDIT FACILITY AND CANCELLATION OF THE EXTENDED CREDIT FACILITY ARRANGEMENT—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the International Monetary Fund and the International Development Association

According to the debt sustainability analysis (DSA) conducted on the basis of the joint International Monetary Fund (IMF)-World Bank debt sustainability framework for low-income countries,[1] the rating of Central African Republic (C.A.R.) is revised from moderate to high risk of debt distress, with the baseline present value (PV) of external debt-to-exports ratio breaching the policy-related threshold during 2013–19 due to a sharp decline in exports in 2013 resulting from the protracted political and security crisis. The baseline debt service-to-exports ratio also breaches the threshold during 2014–16. Medium-and long-term prospects are expected to improve as security improves and the economy recovers from the crisis. Together with relatively moderate levels of highly concessional external debt, the collapse in output and exports are the main drivers of the revision in the assessment, compared with the DSA prepared before the crisis. The DSA shows that C.A.R. is highly vulnerable to terms-of-trade shocks. The analysis further shows that C.A.R. should maintain the minimum concessionality requirement for future borrowing, as this impacts the debt distress rating. Overall, the DSA results suggest that C.A.R. needs to pursue prudent fiscal policies over the medium term while the reconstruction and rebuilding of government institutions demand substantial financial resources, mostly provided by the international development partners.

[1] The LIC DSA compares the evolution over the projection period of debt-burden indicators against policy dependent indicative thresholds, using the three-year average of the World Bank's (WB) Country Policy and Institutional Assessment (CPIA). C.A.R.'s policies and institutions, as measured by the WB's (CPIA), are classified as a "weak performer" with an average rating of 2.74 in 2010–12.

BACKGROUND

1. This analysis is based on the forward-looking debt sustainability framework for low-income countries (LIC DSA).[2] The collapse of the economy resulting from the deterioration of the security conditions at end-2012 and the subsequent humanitarian catastrophe that affected C.A.R., defined the unfavorable macroeconomic conditions underlining the results presented in the baseline of this LIC DSA, as compared to the previous one.[3] The protracted political and security crisis had a devastating impact on the already fragile economy, with output loss estimated at about 36 percent, large current account deficits caused by the fall in exports and the significantly high fiscal financing needs that mark the first years of the DSA. In the medium-and long-run, a return to normalcy is expected to improve export performance; the resumption of international official transfers for budget and investment projects support will contribute to a return to a more sustainable debt path, assuming improved security and availability of highly concessional external financing. C.A.R. growth rate will remain below potential, constrained by weak domestic capacity and a slow recovery in demand. A sharp decline in GDP and exports in 2013 resulting from the crisis would bring the baseline PV of external debt-to-exports and debt service-to-exports ratios above the policy-related threshold from 2013 to 2019 and from 2013 to 2015, respectively. The discount rate has been revised up to 5 percent, and the financing needs of the transitional government are assumed to be covered by highly concessional loans and grants.

2. The accumulation of additional external and domestic payments arrears in 2012 and 2013 constitute the biggest challenge for the sustainability of the public finances and debt in C.A.R. External payments arrears continued accumulating even before the crisis, sometimes for purely technical reason. Most of them are with non-Paris Club creditors and in particular Taiwan Province of China, Libya and China. Most official external payments arrears are inherited from the pre-2009 Heavily Indebted Poor Countries and Multilateral Debt Relief Initiative (HIPC/MDRI) and the authorities 'efforts to reach an agreement with creditors on their settlement were disrupted by the crisis. The authorities signed agreements with most non-Paris Club creditors to negotiate the outstanding debt under conditions similar to the Paris Club, but the process is in the early stages and at least two countries have not responded to the authorities' request. Some additional payments arrears remained on the multilateral debt, in the form of unpaid interest and late interest to the International Fund for Agricultural Development (FIDA) and the European Investment Bank (EIB). On the domestic front, payments arrears were accumulated largely vis-à-vis civil servants and suppliers. The Bank for Central African States (BEAC) is the main domestic creditor.[4]The emergency situation led the government to contract some loans domestically (the Republic of Congo, for

[2] C.A.R. DSA was prepared by the IMF and WB staff in consultation with C.A.R. authorities. The analysis follows the Staff Guidance Note on the Application of the Joint Bank-IMF Debt Sustainability Framework for Low-Income Countries, WB, IMF, 2013.

[3] Updated DSA presented with the Request for a Three-Year Arrangement under the Extended Credit Facility (IMF Country Report No. 12/240).

[4] C.A.R. and Chad are the two CEMAC member countries with outstanding obligations to the BEAC under various facilities. C.A.R. is not servicing its debt to the BEAC.

CFAF 25 billion signed in June 2013)[5], and externally (Angola, US$ 20 million scheduled for the second half of 2014).

3. Total public debt, including domestic arrears, of C.A.R. is estimated at 45.1 percent of GDP as of end-September 2013.[6] External public and publicly-guaranteed debt as of September 2013 amounts to 27.2 percent of GDP, of which non-Paris Club bilateral creditors account for 85.8 percent.[7] Domestic public debt represents 18 percent of GDP. It consists of outstanding credits to the government from domestic commercial banks (6.7 percent), government debt with the BEAC (47.5 percent), and other creditors (45.8 percent).

4. Additional accumulation of domestic arrears in the last quarter of 2013 are estimated to bring the total public debt to 50.8 percent at end-2013. With no final information regarding the additional government arrears from end-September to end-December 2013, an estimation that includes wages, commercial banks, providers and others, would increase the stock of public debt by 5.7 percent of GDP. The estimation is based on the fiscal accounts reported by the government and the consistency checks performed by the Bank and IMF staffs.

Text Table 1. Central African Republic:
Stock of External and Domestic Public and Publicly Guaranteed Debt
(as of September 2013, CFA billions)

	Outstanding[1]	of which Arrears	Share of Outstanding
External Debt	206	126	100.0
Bilateral	177	121	85.8
Paris Club	1	0	0.3
Non-Paris Club	176	118	85.4
Multilateral	29	6	14.2
IDA	12	0	6.0
Other	17	6	8.3
Domestic Debt	136	...	100.0
BEAC	65	...	47.5
Commercial Banks	9	...	6.7
Other	63	...	45.8
Total Public Debt	343	...	

Source: C.A.R. authorities and IMF staff calculations
[1] As of September 2013.

[5] The loan from the Republic of Congo is considered domestic debt because the operation was made through the BEAC (See definition of domestic debt in the TMU).

[6] Public debt includes public and publicly-guaranteed external debt, domestic public debt, budgetary arrears of the central government (including arrears to BEAC), and external and domestic debt of state-owned enterprises.

[7] The main single bilateral creditor on a net present value basis after HIPC/MDRI debt relief is Taiwan Province of China (30 percent of the non-Paris Club debt).

UNDERLYING DSA ASSUMPTIONS

5. The near-term macroeconomic outlook remains influenced by the political and security crisis and the prospects for a recovery with the support of the international community. After significantly declining in 2013, exports, specifically diamonds and wood, are expected to recover and exceed historical levels. Strong donor's support and sustained external demand for wood products and diamonds are expected to help C.A.R. transit out of the crisis. GDP growth will also be sustained at higher than historical rates. Persistently high international oil prices constitute, however, a challenge for the export industries and for the fiscal accounts because the government did not fully implement the pricing mechanism for petroleum products.[8]

6. Exports will slowly resume after the sharp decline in 2013 as international peacekeeping forces reach rural areas and the restrictions under the Kimberly process[9] are lifted. After a significant decline of about 44 percent in dollar value, exports are expected to return to a sustained growth trajectory in 2014–15. The suspension of the Kimberly process is expected to be lifted sometime in late 2014 or early 2015. Exports of wood and cotton would need additional time to be fully restored. Some natural resource exports are expected to start after 2020 with a positive impact on FDI as well.

7. Over the longer term, C.A.R. is expected to achieve steady-state growth of around 6 percent, supported by political stability and sustained public and private investment. A sustained improvement in business confidence based on the return to democratic institutions and pending reforms of legislative, judicial, and administrative systems and, consequently, higher public and private investment should underpin this growth projection (Box 1). Growing exports of a diversified range of primary goods, including gold and cotton, are expected to lift real GDP growth to above 5 percent over the next few years. Consolidation of the political and macroeconomic conditions will drive growth rates to around 6 percent by 2033. Resumption of exports will help improve the external current account deficit over time to around 2½ percent of GDP, with financing initially provided by grants and highly concessional project loans, and the issuance of some government securities. In order to restore debt sustainability, fiscal policy would remain prudent. The authorities' fiscal anchor, the domestic primary balance—which excludes grants and foreign-financed capital spending—would require a significant effort from the government to achieve long-term balance. The primary balance including grants, which determines the debt trajectory more directly, would initially be negative (2014), but will turn to a small surplus in the years after the crisis, helping to stabilize the public debt-to-GDP ratio.

8. There are latent risks to C.A.R.'s macroeconomic outlook. Delays in resolving the domestic political uncertainty and security situation could potentially weigh on donor support and investor confidence. The transition government is tasked with in the return to democratic institutions and failing to

[8] Before the crisis, the government had a mechanism to set petroleum product prices that did not fully pass-through international price adjustments, thus subsidies dragged on fiscal resources.

[9] C.A.R. diamond exports were certified by the Kimberly Process, a joint initiative of governments, industry and civil society that looks into the use of diamonds to finance rebel groups against legitimate governments.

do so would contribute to dim the expectations of a recovery in a post-conflict environment. Adequate investment in infrastructure and improving the business environment are needed to mitigate the damage resulting from the crisis, sustain economic growth and reduce poverty.

Box 1. Central African Republic: Baseline Macroeconomic Assumptions

Real GDP growth: Average annual real GDP growth for 2014–33 is projected at 5.9 percent. This takes into account the expected resolution of the political and security crisis, the return of the displaced people and the resumption of transfers from international donors. The baseline scenario does not take into account potentially higher growth from possible large scale exploitation of natural resources; the high degree of uncertainty surrounding the level of the reserves and the timeline for their exploitation, make it difficult to assess their real long term impact on C.A.R. economy. The relatively high GDP growth assumption is predicated on continued security and political stability and structural transformation of the economy. Sustained economic growth is driven by an increase in private investment, especially in forestry, agriculture, mining, and telecommunications. Public investment is also expected to increase throughout the projection period to reach about 8 percent of GDP per year in 2033, thus improving the infrastructure network that is crucial to boost the agriculture sector, the dominant economic activity. The reestablishment of democratic institutions and the rule of law should prepare the ground for sustained FDI inflows in these sectors. Reconstruction efforts are expected to bear fruit and will place the country in a better position to confront the challenges of inclusive growth in a post-conflict environment.

Inflation: After unusually high inflation in 2013 driven by the scarcity of food and other basic staples, the GDP deflator is projected at 2¼ percent on average for 2014–33; the average CPI inflation in 2014 is expected to reach 4.5 percent and then decline towards the assumed 2 percent rate. The projected inflation rate is in line with the Central African Economic and Monetary Community (CEMAC)'s convergence criterion for CPI inflation of 3 percent.

Current account balance: The current account deficit (including grants) is projected to average 6 percent of GDP for 2014–33. After a sharp decline in exports in 2013, the trade balance is expected to improve over time, driven by a range of factors, including the recovery of the tradable sectors, better competitiveness, and enhanced export performance resulting from structural reforms and the expansion of the infrastructure network to support export diversification; the deficits in the services balance would remain large. The current account deficits would be financed primarily by highly concessional official development assistance (project loans), and some foreign direct investment in natural resources, once the unfavorable environment for investment dissipates.

Government balance: The domestic primary fiscal balance would improve to a surplus of 0.3 percent of GDP by 2018 and then remain around balance in average between 2019 and 2033. The overall fiscal balance (including grants) would average about -¾ percent of GDP for 2014–33, with primary deficits initially lower than needed to stabilize the debt-to-GDP ratio. Domestic revenue is projected to rise from the pre-crisis level of about 11½ percent of GDP (2012), to 20 percent of GDP at end–2033, mainly as a result of steady tax and customs administration improvements and tax policy reform to bring C.A.R. in line with peer countries. Domestic primary expenditures are projected to rise from about 11 percent of GDP in 2012 to 21 percent of GDP in 2033, with most of the increases concentrated in capital spending.

External assistance: After substantial increase in donor's support in 2014 and 2015, grant-equivalent financing is assumed at about 5½ percent of GDP annually in the long run, accounting for 90 percent of total external assistance over the period 2016–33. The grant element of new external loans is expected to be at least 35 percent for the period. Negotiation with non-Paris Club creditors is expected to be concluded on similar terms as for the Paris Club creditors, resulting in the clearance of existing external payment arrears.

Domestic borrowing: In 2012, the government started accessing the securities markets with the issuance of treasury bills. As a result of the crisis, the issued bills have been rolled over with a substantial cost in interest payments. Servicing the emergency loan from the Republic of Congo is also carrying a steep cost in the first post-crisis years. Some rebalancing will be needed to make the burden of domestic debt more sustainable. In the long run, the average real interest rate on domestic currency debt should converge to about 3.

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

9. Under the baseline scenario, the PV of debt-to-exports and the debt service-to-exports ratios are above the thresholds in a protracted manner in the first years (Text Table 2, Figure 1, and Table 1). The PV of debt-to-export ratio exceeds the threshold for the first four years, 2014 to 2019, and the debt service-to-exports ratio remained above the threshold from 2014 to 2017, mainly due to the sharp decline in exports in 2013 and its slow recovery under the baseline scenario. Most generally, the PV of debt ratios are affected in the medium term by the relatively large amount of outstanding debt with non-Paris club creditors. Similarly, debt service-to-revenue ratio starts above the threshold in 2014, but returns to sustainable levels, mostly as a result of the normalization of government operations and revenue mobilization in the post-conflict environment. All debt indicators show stable downward trends over the forecasted period.

Text Table 2. Central African Republic: Policy-Based Thresholds and External Debt Burden Indicators

		Baseline Scenario Ratios		
	Thresholds[1]	2013	2014–33[2]	2014–33
			Proj.	Peak
PV of external debt in percent of:				
GDP	30.0	16.1	12.4	19.1
Exports	100.0	178.2	88.7	225.8
Revenue	200.0	282.9	65.2	296.9
External debt service in percent of:				
Exports	15.0	11.7	6.8	20.8
Revenue	18.0	18.5	4.6	18.5

Sources: C.A.R. Authorities; and IMF and World Bank estimates.

[1] Policy-based thresholds as defined in the LIC DSA framework for a weak policy performer based on the 3-year average CPIA score (2.73 for 2010-2012).

[2] Simple average.

10. Analysis of the impact of additional shocks corroborate the debt sustainability assessment for C.A.R. Volatility in the dollar vis-à-vis the euro makes the country highly vulnerable to exchange rate fluctuations, and affects possible external demand for diamonds and wood. The suspension of the Kimberly certification process for diamonds and the stagnation in timber exploitation had a very damaging impact on the economy. Other non-traditional exports that have potential to initiate an economic diversification process (gold and cotton) have been also stopped due to security problems in the areas of explotation, making it harder for the country to generate foreign currency to service the external debt. A persistent negative shock to exports could exacerbate the risks of debt distress in the future.

11. The vulnerabilities exposed in this DSA call for securing high levels of new concessional loans to finance the fiscal gap generated by the crisis. The appropriate level of concessionality in C.A.R. outstanding external debt, shields the PV of debt-to-GDP ratio from a possible increase in the interest rates

on those loans. This situation can change if the emergency situation forces the government to access financing with high interest rates. The prioritization of government expenditure closely linked to available resources is important to guarantee financial soundness in the medium and long term.

12. The historical scenario could not adequately reflect the baseline prospects for C.A.R. in the near future and the long run (Figure 1 and Table 1.b). This scenario includes the years of high volatility and uncertainty resulting from the conflict, a period in which external financing was constrained by many of the same factors that limited growth. Also, the 2009 debt relief reduced substantially the vulnerabilities of the country vis-à-vis external creditors. The assumptions in the historical scenario are thus substantially different from those embodied in the baseline scenario.

13. The bound tests stress potential risks under an exports shock, but low pre-crisis debt levels of external debt help move debt ratios towards sustainable levels. The most extreme stress test scenario, which assumes either a depreciation of the exchange rate (Figures 1.b, 1.d, and 1.f), an exports shock (Figure 1.c, and 1.e), acts mostly by lowering export values, but does not result in an unsustainable path for the main external debt indicators in the forecasted period. Exports of goods and services have been highly volatile over a low base, and the crisis lowered exports even further. With exports representing only 9 percent of GDP in 2013, an additional shock of one standard deviation on 2014 and 2015 exports would lead to a large and protracted breaching of thresholds on PV of debt-to-exports ratio, but this scenario (Figures 1.c, and 1.e) is also highly unlikely as the baseline scenario already captured the severe impact of the crisis on exports.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

14. Public debt indicators start at a high level in the first projection years as a result of the crisis, but reach sustainable levels once the country returns to normal conditions as depicted in the baseline scenario (Figure 2 and Table 2a). Assuming lower program grants than in the recent past (1 percent of GDP over the long term) for the medium-and long-term and highly concessional financing, all domestic debt indicators are expected to decline gradually. It is further assumed that the government would restore its capacity to sustain prudent fiscal policy by reducing the primary deficit from the first years in the forecast, into a primary balance over the long term; mobilizing higher domestic revenues relative to GDP; and maximizing low-cost financing. Under these assumptions, the PV of public debt-to-GDP ratio could decline by more than 36 percentage points, from 46 percent of GDP in 2011, and stabilize at close to 9.8 percent of GDP in the long run. The PV of public debt-to-revenue (including grants) ratio could fall from 426 percent to 35 percent in the same period.

15. An alternative scenario points to the risk of larger-than-anticipated primary deficits resulting from low revenue mobilization, with a negative impact for the sustainability of the domestic debt. From the different alternative scenarios tested in the LIC DSA for C.A.R., a negative shock affecting government revenues would increase the primary deficits and the PV of public debt-to-GDP ratio, which will remain above the threshold for most of the forecast period (2014 to 2033). In any case, the alternative to maintain the primary balance at the 2013 level is highly unrealistic, as this is precisely the year

of the crisis. Adequate policies to support public finance management efforts and improve domestic revenue mobilization will help avoid the deterioration of debt ratios.

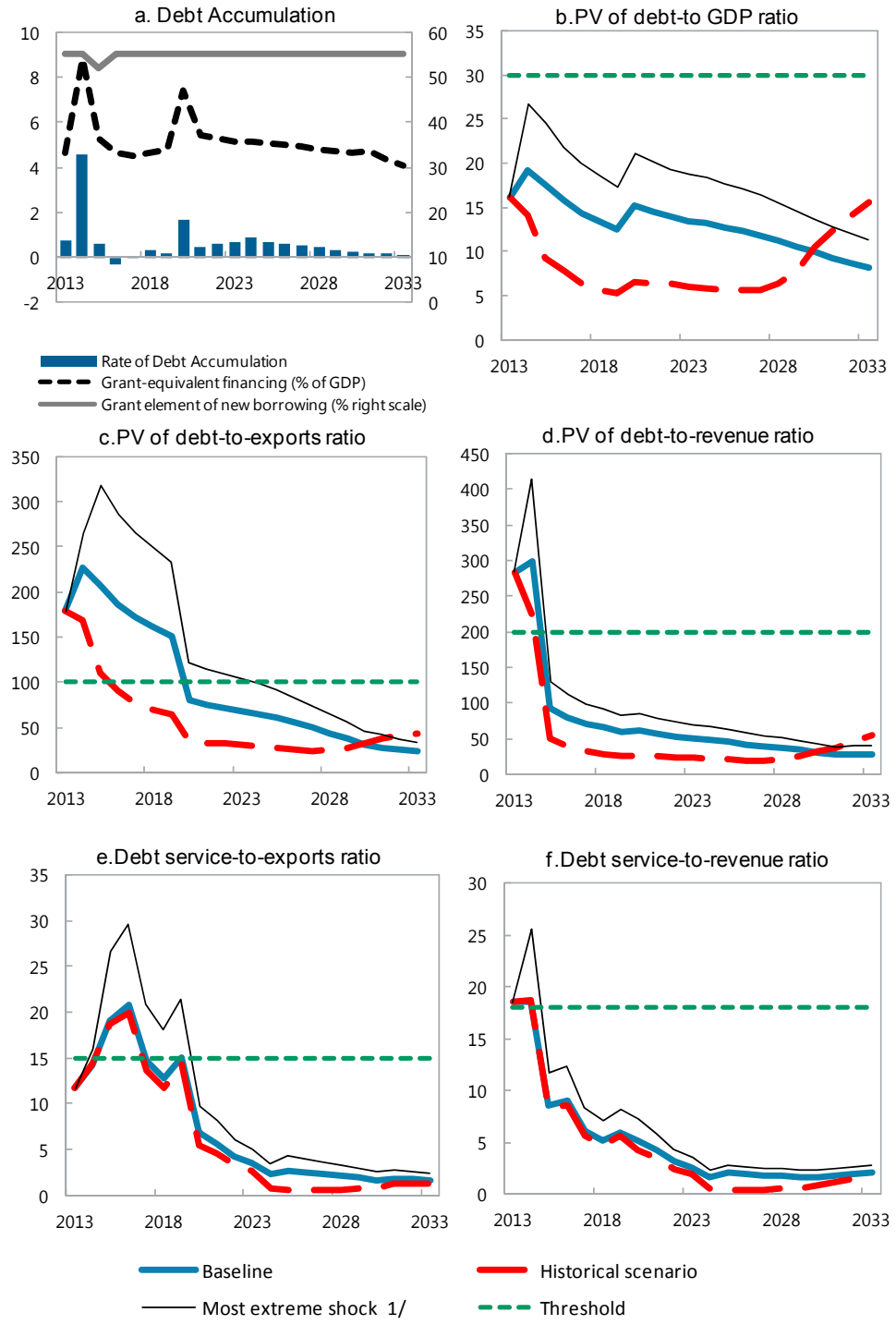
16. All debt indicators would deteriorate further if real GDP growth were to drop temporarily in two consecutive years (Figure 2 and Table 2b). A temporary growth decline—real GDP growth of 3.3 percentage points less than the 2015–16 growth in the baseline—represents the most extreme scenario.[10] In this case, the PV of the debt-to-GDP ratio will end up slightly above the baseline by 2033, the PV of the debt-to-revenue ratio will be 88 instead of 34 percent, and the debt service-to-revenue ratio would remain mostly unchanged. The results suggest that the baseline is very sensitive to the growth assumptions. However, the shocks may be somewhat extreme because the growth standard deviation is high, due to the inclusion of an outlier related to the civil war in 2003. Excluding this outlier would significantly increase the average historical growth and reduce its standard deviation.

DEBT DISTRESS QUALIFICATION AND CONCLUSIONS

17. C.A.R. has a high risk of debt distress, and high loan concessionality is required. The reversal of the substantial improvements achieved following the 2009 HIPC completion point and MDRI debt relief, highlight the vulnerability of the country facing large shocks in output and exports, as well as the associated collapse in fiscal revenues. Accumulation of large domestic payments arrears exposed the difficulties confronted by the government to cover their primary current expenses. The country's debt sustainability exercise points to the fact that C.A.R. is under high risk of debt distress. In addition to the efforts for revenue mobilization in the context of a broken down productive system, the country needs decisive support from development partners to: (i) fill the temporary financing needs of the transition government; and (ii) contribute additional resources to restore democratic institutions and finance a long-term strategy to sustain economic growth, investment and creation of employment and better living standards for the population. This needs to be done without harming debt sustainability, making the recourse to high degree of concessionality a priority for the medium-and long-run. The external debt indicators are particularly sensitive to export growth and terms-of-trade volatility. This suggests that policies to diversify and expand the export base, including improving the business climate for foreign direct investment, reducing transport costs, enhancing productivity, and prudent borrowing are essential for preserving external debt sustainability. The public debt is sensitive to slower GDP growth and an increase in debt-creating flows, confirming that C.A.R. needs to pursue prudent fiscal policies over the medium-term and to consolidate the basis for growth by fostering domestic security, establish the basis for adequate private and public investment, and improve the country's institutional and administrative capacity.

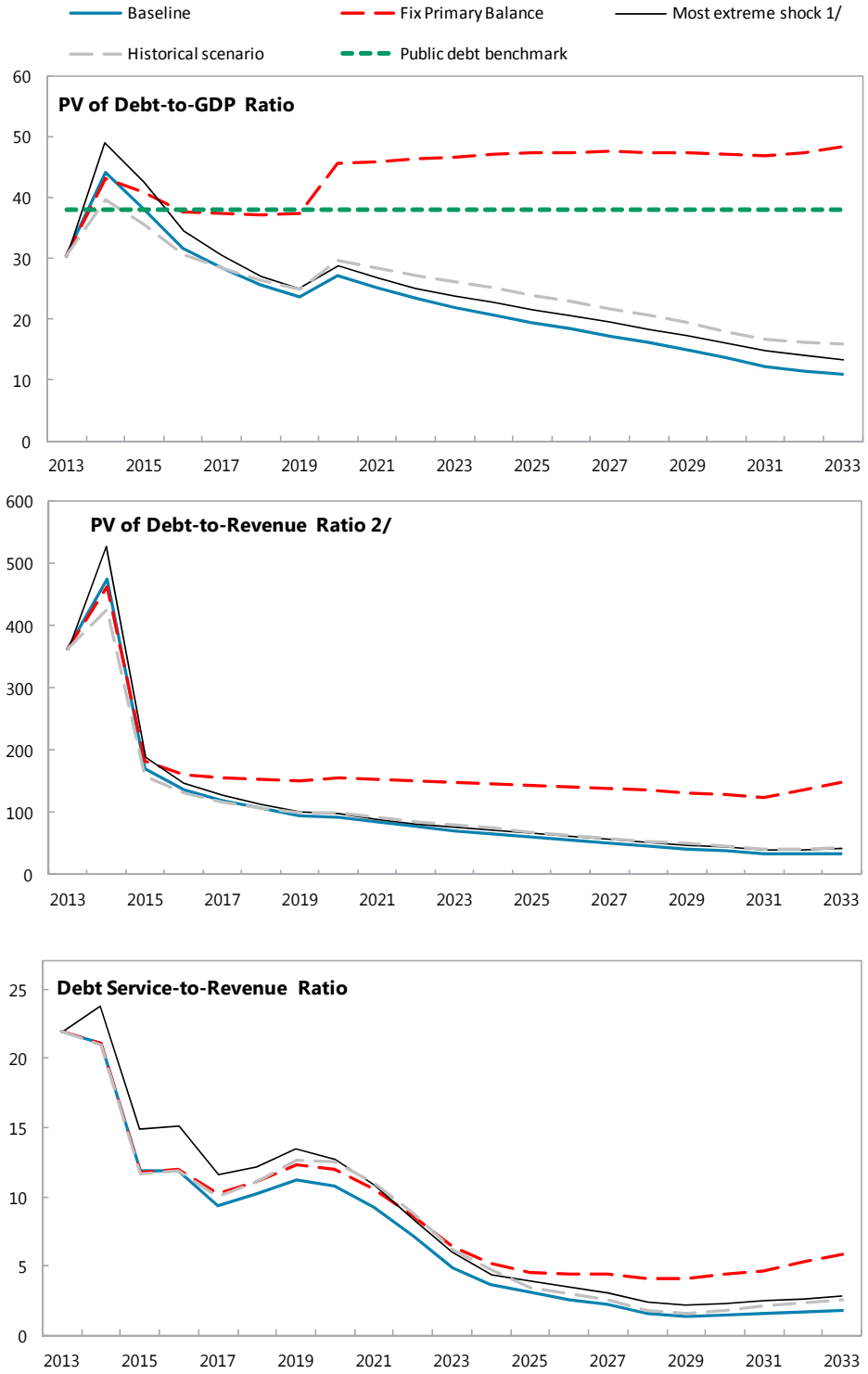
[10] The low growth shock is defined as the average of real GDP growth minus one standard deviation calculated over the 10 years before the 2013 crisis.

Figure 1. Central African Republic: Indicators of Public and Publicly–Guaranteed External Debt Under Alternatives Scenarios, 2013–33[1/]



Sources: C.A.R. Authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figures b. d. and f. it corresponds to a one-time depreciation shock, and; in c. and e. to a shock in exports.

Figure 2. Central African Republic: Indicators of Public Debt under Alternative Scenarios, 2013–33



Sources: C.A.R. Authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Table 1a. Central African Republic: External Debt Sustainability Framework, Baseline Scenario, 2010–33

(in percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**19.8**	**23.2**	**24.6**			**36.6**	**43.3**	**40.1**	**36.3**	**33.3**	**31.0**		**30.8**	**16.2**	
of which: public and publicly guaranteed (PPG)	19.8	23.2	24.6			36.6	43.3	40.1	36.3	33.3	31.0		30.8	16.2	
Change in external debt	3.7	3.4	1.3			12.0	6.8	-3.3	-3.8	-3.0	-2.3		-1.5	-1.1	
Identified net debt-creating flows	7.1	4.0	2.6			22.9	13.1	9.5	1.4	2.0	1.7		1.6	-2.3	
Non-interest current account deficit	**10.0**	**7.4**	**5.5**	**5.5**	**3.6**	**10.5**	**14.2**	**13.7**	**5.6**	**5.8**	**5.3**		**5.3**	**-0.1**	3.2
Deficit in balance of goods and services	14.0	10.3	9.8			15.6	18.0	17.5	9.3	9.5	8.9		12.1	0.9	
Exports	10.6	12.1	11.8			9.0	8.4	8.5	8.5	8.4	8.3		20.0	35.8	
Imports	24.6	22.4	21.6			24.6	26.5	26.0	17.8	17.8	17.2		32.1	36.7	
Net current transfers (negative = inflow)	-4.0	-2.9	-4.3	-3.0	1.8	-5.3	-3.9	-3.8	-3.7	-3.6	-3.5		-6.6	-4.9	-6.0
of which: official	-3.7	-2.2	-3.1			-3.3	-2.1	-2.1	-2.1	-2.1	-2.1		-5.9	-4.6	
Other current account flows (negative = net inflow)	0.0	0.0	-0.1			0.2	0.1	0.0	0.0	0.0	0.0		-0.2	3.9	
Net FDI (negative = inflow)	**-3.1**	**-1.7**	**-3.3**	**-2.4**	**1.7**	**0.0**	**-0.5**	**-2.2**	**-2.1**	**-2.0**	**-1.9**		**-2.0**	**-1.3**	-1.7
Endogenous debt dynamics 2/	**0.2**	**-1.7**	**0.4**			**12.4**	**-0.5**	**-2.1**	**-2.1**	**-1.9**	**-1.8**		**-1.7**	**-0.9**	
Contribution from nominal interest rate	0.2	0.2	0.1			-0.1	0.0	0.0	0.0	0.0	0.0		0.1	0.0	
Contribution from real GDP growth	-0.5	-0.6	-1.0			12.5	-0.5	-2.1	-2.1	-1.9	-1.8		-1.8	-1.0	
Contribution from price and exchange rate changes	0.5	-1.3	1.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-3.3**	**-0.6**	**-1.3**			**-10.9**	**-6.3**	**-12.7**	**-5.2**	**-5.0**	**-4.0**		**-3.1**	**1.3**	
of which: exceptional financing	-3.0	-0.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	10.8			16.1	19.1	17.6	15.7	14.4	13.4		13.5	8.2	
In percent of exports	...	...	91.7			179.0	226.6	207.3	185.9	172.0	161.4		67.3	23.0	
PV of PPG external debt	**...**	**...**	**10.8**			**16.1**	**19.1**	**17.6**	**15.7**	**14.4**	**13.4**		**13.5**	**8.2**	
In percent of exports	**...**	**...**	**91.7**			**179.0**	**226.6**	**207.3**	**185.9**	**172.0**	**161.4**		**67.3**	**23.0**	
In percent of government revenues	**...**	**...**	**93.9**			**284.1**	**297.9**	**93.3**	**80.1**	**71.1**	**65.6**		**49.8**	**28.7**	
Debt service-to-exports ratio (in percent)	**-3.3**	**-22.7**	**-22.1**			**11.7**	**14.1**	**19.0**	**20.8**	**14.7**	**12.7**		**3.5**	**1.6**	
PPG debt service-to-exports ratio (in percent)	**-3.3**	**-22.7**	**-22.1**			**11.7**	**14.1**	**19.0**	**20.8**	**14.7**	**12.7**		**3.5**	**1.6**	
PPG debt service-to-revenue ratio (in percent)	**-3.0**	**-25.4**	**-22.6**			**18.6**	**18.5**	**8.5**	**9.0**	**6.1**	**5.2**		**2.6**	**2.1**	
Total gross financing need (Billions of U.S. dollars)	0.1	0.1	0.0			0.2	0.3	0.2	0.1	0.1	0.1		0.1	-0.1	
Non-interest current account deficit that stabilizes debt ratio	6.3	3.9	4.2			-1.5	7.4	17.0	9.4	8.8	7.6		6.8	0.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.0	3.3	4.1	2.2	3.3	-36.0	1.5	5.3	5.7	5.7	5.8	-2.0	6.1	5.9	6.3
GDP deflator in US dollar terms (change in percent)	-2.7	7.0	-5.0	6.5	8.8	10.7	8.7	6.4	4.1	3.8	3.7	6.2	2.2	1.5	1.2
Effective interest rate (percent) 5/	1.2	1.3	0.6	1.1	0.5	-0.3	-0.1	-0.1	0.0	0.0	0.0	-0.1	0.4	0.1	0.1
Growth of exports of G&S (US dollar terms, in percent)	11.4	26.1	-3.6	6.8	15.5	-45.7	3.2	12.7	9.5	8.6	9.1	-0.4	10.4	11.4	20.4
Growth of imports of G&S (US dollar terms, in percent)	14.3	0.6	-4.4	8.2	12.1	-19.2	18.4	10.0	-24.8	10.1	5.9	0.1	10.5	7.7	14.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	55.3	55.3	52.2	55.3	55.3	55.3	54.8	55.3	55.3	55.3
Government revenues (excluding grants, in percent of GDP)	11.6	10.8	11.5			5.7	6.4	18.9	19.6	20.2	20.5		27.0	28.7	28.6
Aid flows (in Billions of US dollars) 7/	0.1	0.1	0.1			0.1	0.2	0.1	0.1	0.1	0.1		0.2	0.3	
of which: Grants	0.1	0.1	0.1			0.0	0.0	0.1	0.1	0.1	0.1		0.1	0.3	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.1	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			4.6	8.9	5.3	4.6	4.5	4.6		5.2	4.1	5.0
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			75.1	64.7	77.8	86.2	87.7	86.1		87.9	92.7	88.9
Memorandum items:															
Nominal GDP (Billions of US dollars)	2.0	2.2	2.2			1.5	1.7	1.9	2.1	2.3	2.5		3.3	7.5	
Nominal dollar GDP growth	0.2	10.6	-1.1			-29.2	10.3	12.0	10.0	9.7	9.7	3.8	8.4	7.5	7.6
PV of PPG external debt (in Billions of US dollars)			0.2			0.3	0.3	0.3	0.3	0.3	0.3		0.4	0.6	
(PVt-PVt-1)/GDPt-1 (in percent)						0.7	4.6	0.6	-0.3	0.1	0.3	1.0	0.7	0.1	0.5
Gross workers' remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	10.8			16.2	19.2	17.7	15.8	14.4	13.5		13.5	8.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	94.6			184.7	233.9	213.8	191.7	177.1	165.9		68.5	23.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	-22.8			12.1	14.5	19.6	21.5	15.1	13.1		3.6	1.7	

Sources: C.A.R. authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Central African Republic: Sensitivity Analysis for Key indicators of Public and Publicly-Guaranteed External Debt, 2013–33

(percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	16	19	18	16	14	13	**13**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	16	14	9	8	6	6	**6**	16
A2. New public sector loans on less favorable terms in 2013-2033 2	16	22	22	20	19	18	**21**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	16	20	19	17	16	15	**15**	9
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	16	20	19	17	16	15	**15**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	16	21	21	19	18	16	**16**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	16	20	21	19	17	16	**16**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	16	21	24	21	20	18	**18**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	16	27	25	22	20	19	**19**	11
PV of debt-to-exports ratio								
Baseline	179	227	207	186	172	161	**67**	23
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	179	169	110	92	77	69	**30**	43
A2. New public sector loans on less favorable terms in 2013-2033 2	179	266	253	235	224	216	**106**	43
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	179	228	209	187	173	162	**68**	23
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	179	264	318	287	266	249	**103**	34
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	179	228	209	187	173	162	**68**	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	179	243	245	222	206	194	**80**	26
B5. Combination of B1-B4 using one-half standard deviation shocks	179	246	284	257	239	224	**93**	30
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	179	228	209	187	173	162	**68**	23
PV of debt-to-revenue ratio								
Baseline	284	298	93	80	71	66	**50**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	284	222	50	40	32	28	**22**	54
A2. New public sector loans on less favorable terms in 2013-2033 2	284	350	114	101	93	88	**78**	54
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	284	308	103	88	78	72	**55**	32
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	284	307	103	89	79	73	**55**	31
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	284	334	114	98	87	80	**61**	35
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	284	319	110	96	85	79	**59**	32
B5. Combination of B1-B4 using one-half standard deviation shocks	284	333	127	109	98	90	**68**	37
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	284	415	130	112	99	91	**69**	40

Table 1c. Central African Republic: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2013-2033 *(concluded)*

(percent)

Debt service-to-exports ratio								
Baseline	12	14	19	21	15	13	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	12	14	19	20	14	12	**3**	1
A2. New public sector loans on less favorable terms in 2013-2033 2	12	14	21	23	17	16	**5**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	12	14	19	21	15	13	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	12	16	27	30	21	18	**5**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	12	14	19	21	15	13	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	12	14	19	21	15	13	**4**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	12	15	23	25	18	16	**4**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	12	14	19	21	15	13	**3**	2
Debt service-to-revenue ratio								
Baseline	19	19	9	9	6	5	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	19	19	8	9	6	5	**2**	2
A2. New public sector loans on less favorable terms in 2013-2033 2	19	19	9	10	7	6	**4**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	19	19	9	10	7	6	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	19	19	9	9	6	5	**3**	2
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	19	21	10	11	7	6	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	19	19	9	9	6	5	**3**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	19	20	10	11	7	6	**3**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	19	26	12	12	8	7	**4**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	52	52	52	52	52	52	**52**	52

Sources: C.A.R. authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Central African Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–33

(in percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	44.2	47.7	47.5			50.8	68.4	60.5	52.2	47.2	43.2		39.3	18.8	
of which: foreign-currency denominated	19.8	23.2	24.6			36.6	43.3	40.1	36.3	33.3	31.0		30.8	16.2	
Change in public sector debt	4.6	3.5	-0.2			3.3	17.6	-7.9	-8.3	-5.0	-4.1		-2.6	-1.3	
Identified debt-creating flows	0.8	1.3	-3.4			23.5	5.4	-8.3	-3.3	-3.0	-2.3		-2.9	-2.1	
Primary deficit	0.4	1.8	-0.7	-0.4	4.0	6.1	8.6	-1.9	-1.1	-0.6	-0.3	1.8	0.0	-0.3	0.1
Revenue and grants	17.2	13.3	16.4			8.4	9.3	22.5	23.3	23.9	24.2		31.3	32.4	
of which: grants	5.5	2.5	4.9			2.7	2.9	3.6	3.7	3.7	3.7		4.3	3.7	
Primary (noninterest) expenditure	17.6	15.1	15.6			14.5	17.9	20.6	22.2	23.4	23.9		31.4	32.1	
Automatic debt dynamics	0.3	-1.0	-2.8			20.6	-3.2	-6.4	-4.6	-3.7	-3.3		-2.8	-1.3	
Contribution from interest rate/growth differential	-0.9	-1.6	-2.4			22.0	-2.8	-5.7	-3.9	-3.1	-2.9		-2.8	-1.4	
of which: contribution from average real interest rate	0.2	-0.2	-0.5			-4.8	-2.1	-2.3	-0.7	-0.3	-0.3		-0.4	-0.3	
of which: contribution from real GDP growth	-1.2	-1.4	-1.9			26.8	-0.7	-3.4	-3.3	-2.8	-2.6		-2.4	-1.1	
Contribution from real exchange rate depreciation	1.2	0.6	-0.4			-1.3	-0.4	-0.7	-0.7	-0.6	-0.4		...	...	
Other identified debt-creating flows	0.1	0.6	0.1			-3.3	0.0	0.0	2.4	1.3	1.3		-0.1	-0.5	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.1	0.6	-0.1			-3.3	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.3			0.0	0.0	0.0	2.4	1.3	1.3		-0.1	-0.5	
Residual, including asset changes	3.8	2.1	3.2			-20.2	12.2	0.4	-5.1	-2.0	-1.7		0.3	0.9	
Other Sustainability Indicators															
PV of public sector debt	...	...	33.7			30.4	44.2	38.1	31.6	28.3	25.6		22.0	10.9	
of which: foreign-currency denominated	...	...	10.8			16.1	19.1	17.6	15.7	14.4	13.4		13.5	8.2	
of which: external	...	...	10.8			16.1	19.1	17.6	15.7	14.4	13.4		13.5	8.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	0.9	-0.5	-2.2			7.9	10.6	0.8	1.7	1.7	2.2		1.6	0.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	205.9			361.3	474.8	169.2	135.6	118.4	105.9		70.1	33.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	293.2			534.6	688.8	201.8	161.2	140.1	125.1		81.2	37.9	
of which: external 3/	...	...	93.9			284.1	297.9	93.3	80.1	71.1	65.6		49.8	28.7	
Debt service-to-revenue and grants ratio (in percent) 4/	2.7	-17.1	-8.8			22.0	21.1	11.9	11.9	9.4	10.2		4.9	1.8	
Debt service-to-revenue ratio (in percent) 4/	4.0	-21.1	-12.6			32.5	30.6	14.1	14.1	11.1	12.0		5.7	2.0	
Primary deficit that stabilizes the debt-to-GDP ratio	-4.2	-1.7	-0.5			2.8	-9.0	6.0	7.2	4.4	3.8		2.6	1.0	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.0	3.3	4.1	2.2	3.3	-36.0	1.5	5.3	5.7	5.7	5.8	-2.0	6.1	5.9	6.3
Average nominal interest rate on forex debt (in percent)	1.2	1.3	0.6	1.1	0.5	-0.3	-0.1	-0.1	0.0	0.0	0.0	-0.1	0.4	0.1	0.1
Average real interest rate on domestic debt (in percent)	1.6	-0.3	-0.1	-0.4	1.9	-5.9	-2.4	-1.8	0.9	2.4	2.5	-0.7	1.6	0.5	2.7
Real exchange rate depreciation (in percent, + indicates depreciation)	7.8	3.3	-1.9	-1.8	9.9	-3.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.0	2.0	2.7	3.0	1.8	7.2	4.9	4.6	2.2	1.9	2.2	3.9	2.2	1.5	1.1
Growth of real primary spending (deflated by GDP deflator, in percent)	20.5	-11.2	7.8	1.7	8.0	-40.6	25.4	21.0	14.0	11.1	8.2	6.5	8.9	-1.5	8.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	55.3	55.3	52.2	55.3	55.3	55.3	54.8	55.3	55.3	...

Sources: C.A.R. authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Central Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	30	44	38	32	28	26	22	11
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	30	40	36	31	28	26	26	16
A2. Primary balance is unchanged from 2013	30	43	41	38	37	37	47	48
A3. Permanently lower GDP growth 1/	30	45	39	33	30	27	28	29
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	30	45	43	36	34	31	32	28
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	30	42	38	32	29	26	22	11
B3. Combination of B1-B2 using one half standard deviation shocks	30	41	39	33	30	27	27	20
B4. One-time 30 percent real depreciation in 2014	30	48	41	34	30	26	20	10
B5. 10 percent of GDP increase in other debt-creating flows in 2014	30	49	43	35	30	27	24	13
PV of Debt-to-Revenue Ratio 2/								
Baseline	361	475	169	136	118	106	70	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	361	426	158	130	117	107	80	44
A2. Primary balance is unchanged from 2013	361	462	181	162	156	153	149	149
A3. Permanently lower GDP growth 1/	361	477	172	140	124	113	88	89
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	361	484	186	154	138	128	102	85
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	361	449	171	137	120	107	71	34
B3. Combination of B1-B2 using one half standard deviation shocks	361	441	170	138	123	112	84	63
B4. One-time 30 percent real depreciation in 2014	361	517	181	144	124	110	65	30
B5. 10 percent of GDP increase in other debt-creating flows in 2014	361	526	189	148	127	112	76	42
Debt Service-to-Revenue Ratio 2/								
Baseline	22	21	12	12	9	10	5	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	22	21	12	12	10	11	6	3
A2. Primary balance is unchanged from 2013	22	21	12	12	10	11	6	6
A3. Permanently lower GDP growth 1/	22	21	12	12	10	11	5	3
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	22	21	13	13	10	11	6	4
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	22	21	12	12	10	10	5	2
B3. Combination of B1-B2 using one half standard deviation shocks	22	21	12	12	10	11	5	3
B4. One-time 30 percent real depreciation in 2014	22	24	15	15	12	12	6	3
B5. 10 percent of GDP increase in other debt-creating flows in 2014	22	21	12	13	10	11	5	2

Sources: C.A.R. authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.